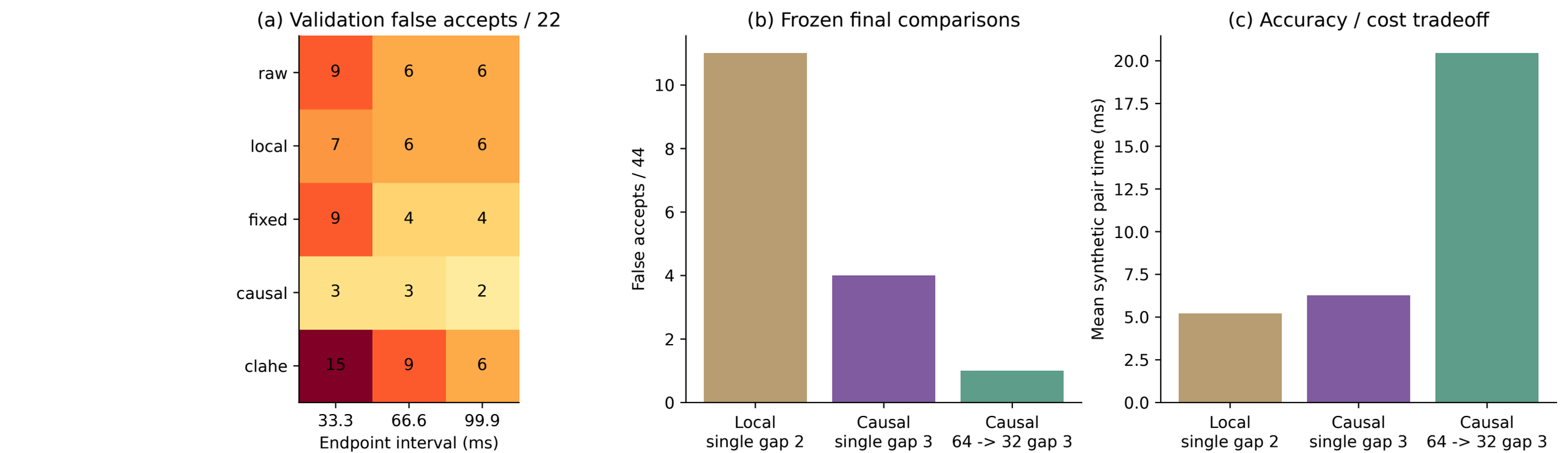

Primary configuration frozen before final: causal EMA alpha=0.08, 48/24, gap 3. Final multipass result is a diagnostic, not used to retune. Midpoint lag = 49.95 ms.